Zhe Yu

Research Assistant | Trustworthy Language Models and Agentic AI

B.Eng. Candidate in Artificial Intelligence | Expected 2027

HKUST (Guangzhou) | LAI Lab · Zhejiang University | IFRC Lab

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Research Profile

I am a B.Eng. candidate in Artificial Intelligence and an undergraduate researcher working on trustworthy language models and agentic AI systems. I study how models internally represent knowledge, evidence provenance, conflict, and risk; why these signals often fail to guide generation and tool use; and how white-box monitoring, audit planning, and verifiable safeguards can improve reliable action and deployment.

Education

Communication University of Zhejiang **2023 – Expected 2027**
B.Eng. in Artificial Intelligence; undergraduate advisor: Dr. Hao Zeng *Hangzhou, China*

Research Experience

The Hong Kong University of Science and Technology (Guangzhou) **May 2026 – Present**
Research Assistant, LAI Lab; advised by Prof. Chengwei Qin *Guangzhou, China*

- At HKUST(GZ), research evaluator reliability, evidence attribution, audit planning, and safeguards for trustworthy language models and agentic AI.

Binjiang Institute of Zhejiang University **Nov. 2025 – Present**
Research Intern, IFRC Lab; supervised by Dr. Meng Han and Dr. Wenpeng Xing *Hangzhou, China*

- Participate in the Guangdong Provincial Key R&D Program *Safety System Research and Applications for Multimodal Large Models*, spanning trustworthy LLMs, RAG faithfulness, white-box monitoring, agentic AI safety, and medical-LLM safety.
- Participate in the National Key R&D Program Young Scientists Project *A New Trust Framework Based on Blockchain*, focusing on verifiable model ownership and on-chain/off-chain trust.

University of Malaya **Jan. 2025 – Feb. 2025**
Visiting Student *Kuala Lumpur, Malaysia*

Westlake University **Mar. 2024 – Sep. 2024**
Visiting Student; supervised by Dr. Ziyang Zhang *Hangzhou, China*

Selected Research Contributions

- **Retrieval grounding and hallucination detection.** Led problem formulation, method design, experiments, and writing for DISF, a dual-path white-box framework that models Conflict, Drift, and Instability in retrieval-augmented generation; accepted at ACL 2026.
- **Evidence attribution and intervention.** Developed internal-signal methods for retrieval-memory conflict, including FIDES for token-level training-free intervention and the Attribution Blind Spot for distinguishing parametric-memory and retrieved-evidence routes behind identical outputs.
- **Audit and evaluator reliability.** Extended hallucination scores into parent-preserving audit plans with PLAIT and studied why item calibration cannot certify experimental contrasts with GAVEL's paired audits and pair-level correction.
- **Representation-action dissociation.** Led mechanistic studies showing that source and conflict information can be linearly available before it influences tool choice or answer policy, including Knowing Is Not Acting and Whose Thoughts?; both are under review at NeurIPS 2026.
- **Monitoring and verifiable deployment.** Led LatentAudit and RETINA-SAFE / ECRT for real-time white-box monitoring and high-risk medical triage; contributed to privacy-preserving ownership verification in ZK-FPE, accepted at ACM TURC 2026.

Selected Peer-Reviewed Publications

1. **Zhe Yu***, Wenpeng Xing*, Wenjie Luo, Weize Xu, Lingtong Huang, Yourong Chen, Changting Lin, Meng Han. *DISF: Detecting Hallucinations in Retrieval-Augmented Generation via Dual-path Internal State Forcing Framework.* **ACL 2026**. [PDF]
2. Zhiguo Ma*, **Zhe Yu**, Wenpeng Xing, Yourong Chen, Meng Han. *ZK-FPE: Blockchain-Verifiable Model Fingerprinting with Zero-Knowledge Privacy for Ownership Attribution.* **ACM TURC 2026**. [PDF]
3. Weiping Yu, Weihan Wang, Mingyuan Yan, Keyang He, **Zhe Yu**, Wenpeng Xing, Liyuan Liu, Meng Han. *Scalable and Trusted Metadata-Coordinated Tiered Off-Chain Storage with Dynamic On-Chain Mapping for Recovery-Safe and Low-Latency IoT Data Management. Electronics*, 2026, 15, 2806. [DOI]

Manuscripts Under Review

1. **Zhe Yu**, Yunzhao Wei, Wenpeng Xing, Wenjie Luo, Zaobo He, Quan Chen, Meng Han. *PLAIT: From Hallucination Scores to Parent-Preserving Audit Plans*. Under review at **AAAI 2027**.
2. **Zhe Yu***, Wenpeng Xing*, Bo Yang, Chen Ye, Gaolei Li, Yunzhao Wei, Meng Han. *The Attribution Blind Spot: Detecting When Language Models Rely on Memory Rather Than Retrieved Context*. Under review at **AAAI 2027**. [arXiv]
3. **Zhe Yu***, Wenpeng Xing*, Chen Ye, Xuyang Teng, Bo Yang, Changting Lin, Meng Han. *Calibration Does Not Certify a Contrast: Identification and Correction for LLM Evaluators*. Under review at **AAAI 2027**.
4. **Zhe Yu**, Wenpeng Xing, Zhenhua Xu, Xingxing Yang, Meng Han. *Knowing Is Not Acting: Representation–Action Dissociation in Indirect Prompt Injection*. Under review at **NeurIPS 2026**.
5. **Zhe Yu**, Wenpeng Xing, Zhenhua Xu, Ruiqi Zhang, Meng Han. *Whose Thoughts? Chain-of-Thought Override in Reasoning-Tuned Language Models*. Under review at **NeurIPS 2026**.
6. **Zhe Yu***, Wenpeng Xing*, Tiancheng Zhao, Mohan Li, Changting Lin, Meng Han. *FIDES: Faithful Inference via Deep Evidence Signals for Retrieval-Memory Conflict in RAG*. Under review at **EMNLP 2026**.
7. **Zhe Yu***, Wenpeng Xing*, Gaolei Li, Shuguang Xiong, Hongzhi Wang, Xuyang Teng, Meng Han. *Cordon-MAS: Defending RAG against Knowledge Poisoning via Information-Flow Control*. Under review at **EMNLP 2026**. [arXiv]
8. **Zhe Yu***, Wenpeng Xing*, Yunzhao Wei, Hongzhi Wang, Xuyang Teng, Meng Han. *Composition Collapse: Stable Factual Knowledge Does Not Imply Compositional Reasoning*. Under review at **EMNLP 2026**. [arXiv]
9. Zhenhua Xu, Qichen Liu, Zhebo Wang, **Zhe Yu**, Xixiang Zhao, Wenpeng Xing, Dezhang Kong, Mohan Li, Meng Han. *Fingerprint Vector: Enabling Scalable and Efficient Model Fingerprint Transfer via Vector Addition*. Under review at **EMNLP 2026**.
10. **Zhe Yu***, Wenpeng Xing*, Meng Han. *LatentAudit: Real-Time White-Box Faithfulness Monitoring for Retrieval-Augmented Generation with Verifiable Deployment*. Under review at CoLM 2026. [arXiv]
11. **Zhe Yu***, Wenpeng Xing*, Meng Han. *From Retinal Evidence to Safe Decisions: RETINA-SAFE and ECRT for Hallucination Risk Triage in Medical LLMs*. Under review at MICCAI 2026. [arXiv]

* Equal contribution. Review statuses are a snapshot as of August 2026.

Earlier Publications

1. F. Zhou, C. Chang, Q. Chang, H. Zhang, **Zhe Yu**, W. Liu, J. Li, J. Yang. *Orthogonal salinity and temperature detection via paralleled dual all-fiber interferometers. Optics Communications*, 583 (2025): 131688. [DOI]
2. **Zhe Yu**, H. Zeng, Y. Zhao, X. Zhang, Z. Wang, Y. Tao, M. Yuan, X. Sun. *Bibliometric analysis of physical education research in China from 2014 to 2024*. ACM ICETM, 2025, pp. 128–132. [DOI]

Patents

1. **Zhe Yu**, Wenpeng Xing, Meng Han. *A hallucination detection method based on dual-path internal state forcing logic for retrieval-augmented generation in large language models*. Chinese Patent Application No. 202610260408X. Under review.
2. Meng Han, **Zhe Yu**, Jiayan Hu, Rongchang Li, Wenpeng Xing, Jingyi Yu, Zhen Hong, et al. *A post-processing method, system, device, and medium for hallucination detection in large language models based on adaptive order statistics aggregation*. Chinese Patent Application No. 2026107898102, filed June 3, 2026. Pending.
3. **Zhe Yu**, Jiayan Hu, Jingyi Yu, Weihang Yu, Wenpeng Xing, Jing Xiong, Yourong Chen, Zhen Hong, et al. *A hallucination detection method, system, and device for large language models based on multi-dimensional heterogeneous feature fusion*. Chinese Patent Application No. 2026107899270, filed June 3, 2026. Pending.

Industry Experience

BoostEngine **Jul. 2025 – Oct. 2025**

Hangzhou, China

R&D Intern, Full-Stack Development / AI Agents *Remote with New York team*

- Built a multi-brand Manifest V3 Chrome extension for TikTok Shop operations across 20 regional domains and 9 language packs using React, TypeScript, Vite, and Chrome APIs.
- Developed Spring Boot analytics services for 8,000+ creator collaborations and built a FastAPI / Redis / MySQL synchronization system with anti-loop control, queue coalescing, monitoring, and alerts.

Awards and Technical Skills

- **Silver Award**, CCB Cup Zhejiang Provincial International College Students' Innovation Competition, 2024, for the project *Scoliosis Detection AI System*.
- **Research:** Python, PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, FAISS, vLLM, W&B, MATLAB.
- **Engineering:** TypeScript / JavaScript, Java, React, FastAPI, Spring Boot, SQLAlchemy, MyBatis-Plus, MySQL, Redis, Docker, Chrome Extensions.